Nils Kuhn

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EDUCATION

Stanford University Stanford, CA — Sep 2024 – Jun 2026
M.S. Electrical Engineering
- **GPA 4.0/4.0** | Stanford Tuition Fellowship | DAAD scholarship | Friedrich-Ebert-Stiftung scholarship (B.S. and M.S.)
- **Graduate coursework:** Probability and Statistical Inference (EE 278), Stochastic Calculus and Control (MS&E 322), Convex Optimization (EE 364A), Reinforcement Learning (CS 234), Deep Reinforcement Learning (CS 224R), Machine Learning (CS 229)

Leibniz University Hannover Hannover, Germany
B.S. Mechatronics, graduated with distinction — Oct 2021 – Sep 2024
- **Grade 1.2** (German scale, 1.0 = best) | **Jürgen Ulderup Award** (1 of 6 graduates of the faculty, €1,500)

B.S. Engineering and Business Administration, graduated with distinction — Oct 2022 – Aug 2024
- **Grade 1.2** (German scale, 1.0 = best) | **Wilhelm Launhardt Award** (1 of 5 graduates of the faculty)
- **Coursework:** Finance and Capital Markets, Capital Market Theory: Derivatives, Game Theory, Microeconomic Theory

RESEARCH EXPERIENCE

Meta FAIR Core Learning and Reasoning — Jul 2026 – Present
Research Scientist 2
- Research **world models** that learn from video to enable zero-shot generalization in robot planning and control.
- Pre-train, train and evaluate large models with **distributed training across multiple GPU nodes** in PyTorch and JAX.

Stanford AI Lab Interactive Perception and Robot Learning Lab — Sep 2025 – Jun 2026
Research Assistant, advised by Prof. Jeannette Bohg | Co-author, CoRL 2026 [2]
- Co-developed FACT, which traces failures of flow-matching VLA policies in contact-rich manipulation to **two causally distinct modes** and fixes each with a targeted mechanism, validated through **ablations**.
- Co-designed time-aware force injection for **sparse force/torque time series**: a shared **causal TCN** encodes 2 s of history, with gradients gated to contact events.
- FACT reached **66% average success vs. 41%** for the best prior baseline across 5 tasks and **about 2,500 real-world rollouts**, with significance tested by **Fisher's exact tests**.

European Space Agency — Jun 2025 – Sep 2025
AI for Space Operations Intern | First author, Acta Astronautica 2026 [1]
- Built a validation-gated **continual learning** framework for **anomaly detection** with **predictive models** on **noisy, nonstationary, multivariate** satellite telemetry, deploying an update only if it improves on held-out data.
- Benchmarked 6 continual learning and 5 active learning methods on the ESA-ADB benchmark. Tuned hyperparameters in-sample on the first half of a mission and evaluated them **out-of-sample** on the second half and on a second mission.
- Reached **92.9% event-wise $F_{0.5}$ vs. 78.6%** for the static baseline with **under 5% of its training data**, and cut the run-to-run std of forecast error by **over 90%**.

PUBLICATIONS

[1] **Kuhn, N.**, Sánchez Gómez, B., Moreno Blasco, N., Caserman, P., and Antonello, F. Validation-Gated Continual Learning for Anomaly Detection in Satellite Telemetry. *Acta Astronautica*, vol. 249, pp. 971–985, 2026. [Paper]

[2] Parés-Morlans, C., **Kuhn, N.**, Liu, I., Longhini, A., and Bohg, J. Demystifying When and Why VLAs Fail in Contact-Rich Tasks and How to Fix Them. *Conference on Robot Learning (CoRL)*, 2026, to appear. [arXiv]

SELECTED PROJECTS

Continuous-Time Diffusion Policies [Report] — Sep 2025 – Dec 2025
Stochastic Calculus and Control (MS&E 322)
- Reformulated diffusion policies as **controlled SDEs** (VP, VE and critically damped Langevin) and implemented Euler, Euler-Maruyama and Heun samplers.
- VP-SDE with the Euler-Maruyama solver reached **0.78 to 0.80 success** on the PushT visuomotor benchmark.

RL Trading in Low-Liquidity Markets [Report] — Jan 2025 – Mar 2025
Reinforcement Learning (CS 234)
- **Designed and implemented three action-space formulations** for a daily-trading **MDP**, two with a **risk-neutral** and one with a **risk-averse reward**, and ran the parameter search for DQN, Double DQN and Transformer Q-network agents.
- Evaluated each agent **out-of-sample** over **11 seeds** on the final year of the First North Growth Market index and one low-volume Swedish stock, where Double DQN matched buy-and-hold **Sortino (1.52 vs. 1.53)**.

SKILLS

Programming Python, C++, C
Frameworks and Tools PyTorch, PyTorch Distributed, JAX, NumPy, Pandas, CVXPY, Git, AWS, ROS, LaTeX